A Regime-Aware Explainable Machine Learning Framework for Energy Prediction and Anomaly Detection in Smart Buildings

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***Abstract*—Energy use in a smart building varies with schedules, occupancy, and operating conditions throughout the day. A single fixed anomaly threshold can therefore raise too many alerts or overlook unusual behavior. This study evaluates a regime-aware framework that links energy forecasting with residual-based anomaly detection and model explanations. A LightGBM gradient-boosted tree model predicts expected energy from time-based and lagged features constructed without using future observations. The absolute difference between observed and predicted energy is assessed with global, rolling, and regime-specific thresholds. For each inferred operating regime, the detector keeps a history of past residuals and uses that history to update its threshold. Experiments were conducted on a deterministic 30-building subset of the annotated LEAD1.0 commercial-building benchmark using a chronological 70/15/15 split. On the held-out test period, the predictor achieved an MAE of 6.005 and RMSE of 12.818, compared with 7.363 and 16.615 for persistence forecasting. The proposed detector achieved 37.2% precision, 12.8% recall, an F1-score of 0.190, a PR-AUC of 0.181, and a ROC-AUC of 0.722, while reducing false alarms to 6.49 per 1,000 normal observations compared with 10.23 for the global residual threshold. These results suggest that regime-aware thresholding can improve alert precision and reduce false alarms, but the lower recall shows that some anomalies are missed. Feature attributions are explanations of model behavior, not proof of a physical fault.**

***Index Terms*—Smart buildings, building energy, machine learning, anomaly detection, residual analysis, adaptive threshold, explainable artificial intelligence, SHAP, time-series analytics.**

I. Introduction

Smart meters and building-management systems record energy use at regular intervals. These records can help operators forecast demand, monitor equipment, and spot unusual consumption. Interpreting them is not straightforward because expected energy use changes with the hour, occupancy, weather, schedules, and equipment state. A reading that looks unusual overnight may be normal during a busy period. A detector that relies on one fixed limit may therefore send unnecessary alerts or fail to flag a meaningful deviation.

Recent research has combined data engineering and machine learning to forecast building energy use and identify possible faults. Sekhara et al. described a cloud-based pipeline that paired forecasting with dynamic Bollinger-band detection and reported strong forecasting results from XGBoost; they also discussed the difficulty of working with limited fault labels and changing operating conditions [1]. A review of predictive maintenance in smart buildings further identifies data quality, domain shift, limited labels, interoperability, validation, and real-time deployment as continuing challenges [2].

Several studies use deeper models to capture more complex patterns. Examples include context-aware Transformers for supervised anomaly detection [4]; Bi-LSTM and hybrid CNN-Bi-LSTM models have been used for residential energy prediction and anomaly detection [5]; and TCN, XGBoost, and VAE components have been combined in building energy-management systems [6]. Neural architecture search and variational autoencoders have also been explored for time-series anomaly detection and transfer learning [7], [8]. Other recent work has examined context-focused explanations for energy anomalies [14], explainable forecasting and dynamic anomaly detection in smart campuses [15], [16], and machine-learning-based prediction and anomaly detection for smart homes [17]. These methods can be useful, although their added complexity may not be necessary when the goal is practical and interpretable monitoring.

This study asks the following question: *Can a lightweight prediction-residual detector with regime-aware adaptive thresholds reduce false alarms while maintaining useful anomaly-detection performance under changing building operating conditions?* The framework also investigates whether model-level feature attribution can provide practical context for detected deviations.

This paper makes four practical contributions. First, it evaluates a pipeline that connects energy prediction to anomaly detection. Second, it uses prediction residuals to identify unusual observations. Third, it adapts the decision threshold to the inferred operating regime and reports a normalized

TABLE I: Research Hypotheses and Evaluation Focus

ID	Evaluation focus
H1	Prediction residuals provide a useful anomaly signal.
H2	Regime-aware thresholds reduce false alarms relative to a global threshold.
H3	Feature attribution provides useful model-level context.
H4	LightGBM training time is practical in the tested environment.

deviation score. Fourth, it evaluates forecasting error, detection quality, false-alarm burden, and model explanations separately. The contribution is the tested combination and its evaluation; the individual components are established techniques rather than newly invented algorithms.

II. Problem Statement and Research Gap

Prior work shows that machine-learning models can forecast energy demand and identify unusual patterns, but several practical questions remain. Residuals may have different distributions during different operating regimes, so one global threshold may not work equally well at all times. Fault labels are also limited in many building datasets, making reliable evaluation difficult. More complex neural models can increase computational cost, while a simple binary alert may give an operator little information about why an observation was flagged. Forecasting errors provide a natural starting point because they measure the difference between expected and observed energy use.

This study evaluates a compact pipeline in which forecasting, residual analysis, regime-specific thresholding, and feature attribution are assessed together. We do not argue that anomaly detection or explainable machine learning is missing from the building-energy literature. Instead, we test whether this combination provides a measurable benefit when evaluation follows time order and operating conditions may change.

A. Novelty and Scope of the Contribution

The novelty claim is limited. Forecasting, residual monitoring, adaptive thresholds, and feature attribution have all been studied previously. This work evaluates a specific sequence: predict expected energy from past-only features, calculate the residual, adjust the threshold using the current operating regime, and record the deviation alongside model-level feature attributions. The experiments test whether this combination changes detection behavior in a useful way; they do not claim that any individual technique is new.

III. Literature Survey

Prior studies have explored interpretable energy prediction in industrial buildings, dimensionality reduction for building-energy forecasting, indoor-temperature prediction using IoT data, and statistical anomaly detection for smart premises [9]–[12]. Table II positions representative forecasting and anomaly-detection studies relative to the present work. The comparison is qualitative because the studies use different datasets, labels, and evaluation protocols.

IV. Proposed Methodology

The framework processes each observation in stages, beginning with the energy record and ending with an anomaly event that can be inspected. Forecasting and the anomaly decision are kept separate so their performance can be measured independently.

A. Data Preprocessing

All observations are ordered by timestamp before features are created. Duplicate rows are removed, missing values are handled using statistics calculated from the training period, and invalid readings are checked. Preprocessing decisions are not fitted on the test period, helping prevent future information from influencing the model.

B. Feature Engineering

The model uses time-based features such as hour, day of week, weekend status, and month, together with past energy readings and rolling summaries. Weather or occupancy variables can be included when available. Each lag and rolling feature is calculated only from observations that would already be available at prediction time.

C. Energy Prediction

The primary predictor is a gradient-boosted decision-tree model implemented with LightGBM. For observation t,

$$\hat{y}_t = f(x_t), \tag{1}$$

where y_t is observed energy and $\hat{y}_t$ is predicted energy. A persistence baseline using the previous hourly reading is included to contextualize forecasting accuracy. Forecasting performance is measured using

$$MAE = \frac{1}{N}\sum_{t=1}^{N} |y_t - \hat{y}_t| \tag{2}$$

and

$$RMSE = \sqrt{\frac{1}{N}\sum_{t=1}^{N} (y_t - \hat{y}_t)^2}. \tag{3}$$

D. Residual Signal

The signed residual is

$$e_t = y_t - \hat{y}_t, \tag{4}$$

and its magnitude is

$$r_t = |e_t|. \tag{5}$$

A large residual means that the observed reading differs substantially from the model's estimate. It is a signal worth investigating, but it does not by itself establish that equipment has failed.

TABLE II: Positioning Against Representative Recent Studies

Study	Main method	Primary strength	Relation to proposed framework
Sekhara et al. [1]	XGBoost prediction + dynamic Bollinger Bands	Integrated data engineering and ML pipeline	Closest base work; this study focuses on residual decision adaptation and controlled false-alarm evaluation.
Seknametla et al. [3]	LightGBM + Isolation Forest	Forecast/anomaly integration for peak demand	Uses prediction residual as the primary anomaly signal.
Almadani et al. [4]	Context-gated Transformer	Supervised contextual anomaly detection	This study emphasizes a lightweight decision layer.
Hamayat et al. [5]	Bi-LSTM/CNN-Bi-LSTM + moving average	Deep forecasting and anomaly detection	This study targets a simpler deployment approach.
Vizhiyarasi et al. [6]	TCN + XGBoost + VAE	Multi-component BEMS architecture	The present work isolates adaptive residual detection.
Dissem et al. [7]	Reinforcement-learning NAS for autoencoders	Automated representation design	The present work uses explicit residual decision logic.
Shahid et al. [8]	VAE + transfer learning	Cross-domain anomaly learning	The present work emphasizes local regime adaptation.

E. Operating-Regime Identification

Building loads follow different patterns across workdays, weekends, seasons, and operating states. A lightweight clustering step groups observations using normalized temporal and rolling features. For each group, the detector keeps a history of residuals observed before the current time step, so the current observation does not influence its own threshold.

F. Adaptive Residual Threshold

The global baseline is

$$\tau_{global} = Q_q(R_{train}), \tag{6}$$

where Q_q is a high residual quantile. The proposed regime-aware threshold is

$$\tau_t = Q_q(R_k(t)) + \lambda IQR(R_k(t)), \tag{7}$$

where IQR is the interquartile range and λ controls tolerance to local variability. An anomaly is declared when

$$a_t = \begin{cases} 1, & r_t > \tau_t, \\ 0, & r_t \le \tau_t. \end{cases} \tag{8}$$

For the reported experiment, the global and rolling thresholds use the 0.995 residual quantile. The proposed regime-aware threshold uses the 0.99 quantile plus 0.25 times the regime-specific IQR, with a 1,000-observation per-regime residual history. These settings were fixed before evaluation on the held-out test period.

G. Severity and Explainability

A normalized severity score is defined as

$$S_t = \frac{r_t}{\tau_t + \epsilon}. \tag{9}$$

A detected event can be reported with timestamp, observed energy, expected energy, residual, threshold, severity, and top contributing features. SHAP, a game-theoretic approach for explaining model predictions, or a compatible attribution method is used to explain the prediction [18]. The attribution is a model-level explanation and is not treated as proof of physical causality.

V. System Architecture

Figure 1 summarizes the processing stages and shows how prediction, regime identification, thresholding, and explanation connect.

A. Operational Decision Record

Each detected event is represented by a compact record: $(timestamp, building, y_t, \hat{y}_t, r_t, \tau_t, a_t, S_t, E_t)$, where E_t contains the highest-contributing explanatory features. This representation supports both quantitative evaluation and dashboard visualization.

VI. Experimental Design

A. Benchmark and Chronological Split

Experiments use LEAD1.0, an annotated commercial-building electricity dataset introduced for large-scale energy anomaly detection [13]. The supplied file contains hourly readings and anomaly labels for 200 buildings during 2016. For a controlled, reproducible experiment, the first 30 building identifiers after sorting were selected. After observations with missing target values were removed and leakage-safe lagged and rolling features were constructed, 198,030 observations remained. The data were divided chronologically into 70% training, 15% validation, and 15% test periods. No random shuffling was used.

B. Baselines

Four detection configurations are evaluated: M1 global residual threshold; M2 rolling residual threshold without explicit regimes; M3 Isolation Forest, an established tree-based outlier-detection method [19]; and M4 the proposed regime-aware adaptive residual threshold. The same LightGBM predictor supplies residuals for M1, M2, and M4, helping isolate the effect of the anomaly-decision rule from changes in the forecasting model.

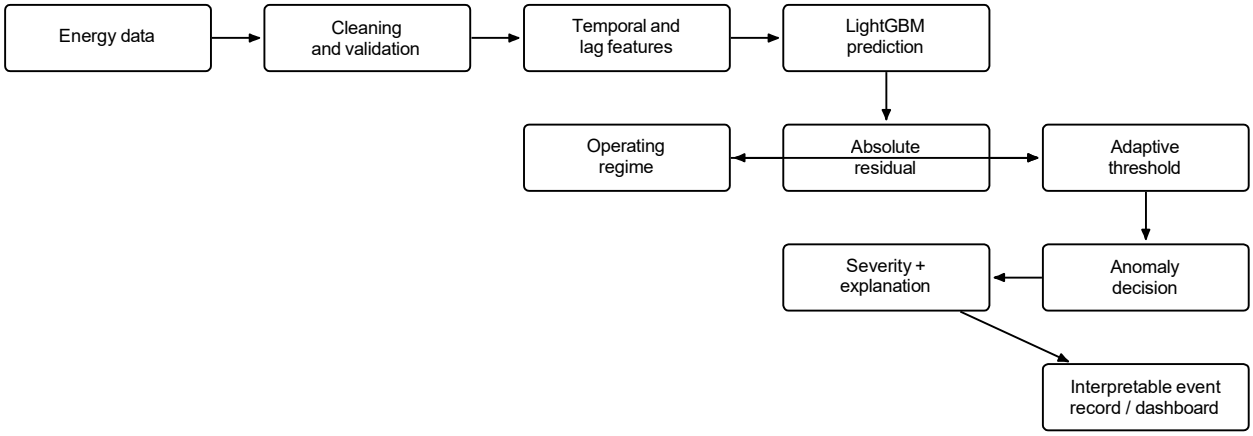


Fig. 1: Regime-aware adaptive residual framework used in the experiment.

C. Metrics

Detection performance is evaluated using precision, recall, F1-score, and PR-AUC:

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}, \tag{10}$$

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}. \tag{11}$$

False alarms per 1,000 normal observations are also reported because operator workload is an important deployment consideration. In this experiment, only model-training time was measured; inference latency and model size were not evaluated.

D. Evaluation Configurations

The evaluated configurations are: A1, the predictor with a global threshold; A2, the predictor with a rolling threshold; A3, the predictor with a regime-aware threshold; and A4, the regime-aware detector accompanied by feature explanations. Separate quantitative ablation results were not calculated for the explanation component. Therefore, explainability is treated as an inspection aid, and no claim is made that it improves detection metrics.

VII. Results and Discussion

The experiment was executed on 198,030 usable observations from 30 buildings. The test set contains 29,698 observations, of which 2.92% are labeled anomalous. All detection thresholds were computed from information available before each test observation; anomaly labels were not used to update the residual histories.

A. Forecasting Performance

On the test period, LightGBM recorded an MAE of 6.005 and an RMSE of 12.818. The persistence baseline, which uses the previous hourly reading as its prediction, recorded an MAE of 7.363 and an RMSE of 16.615. These values correspond to reductions of 18.44% in MAE and 22.86% in RMSE. For this subset, the learned predictor therefore estimated expected energy more accurately than the persistence baseline.

TABLE III: Forecasting Performance on the Held-Out Test Period

Model	MAE	RMSE
Persistence (lag-1)	7.363	16.615
LightGBM predictor	**6.005**	**12.818**

TABLE IV: Anomaly Detection Performance on the Held-Out Test Set

Method	Prec.	Rec.	F1	PR-AUC	ROC-AUC	FP/1000
Global residual	0.293	0.141	0.190	0.152	0.697	10.23
Rolling residual	0.220	0.075	0.112	0.091	0.655	8.01
Isolation Forest	0.301	**0.189**	**0.232**	**0.193**	0.682	13.22
Proposed adaptive	**0.372**	0.128	0.190	0.181	**0.722**	**6.49**

B. Anomaly Detection Performance

Table IV shows that the proposed method achieved the highest precision among all compared methods (0.372) and the lowest false-alarm rate (6.49 per 1,000 normal observations). This is a 36.56% reduction relative to the global residual threshold, based on the reported rounded rates. Its ROC-AUC was 0.722, compared with 0.697 for the global threshold, 0.655 for the rolling threshold, and 0.682 for Isolation Forest. The trade-off is lower recall than Isolation Forest: the proposed method misses more labeled anomalies. Its F1-score is 0.190, effectively the same as the global threshold after rounding. The clearest measured advantage is better precision and fewer false alarms, not an improvement across every metric.

C. Residual and Threshold Behavior

In Fig. 3, the adaptive threshold changes over time rather than staying at one global value. It responds to the residual history for the inferred regime: more variable regimes can receive a wider boundary, while more stable regimes can receive a tighter one. The test results are consistent with this design reducing false alarms. At the same time, lower recall indicates that some less pronounced anomalies may remain undetected.

D. Explainability Analysis

We calculated SHAP values for a random sample of 1,000 test observations to examine which features most influenced

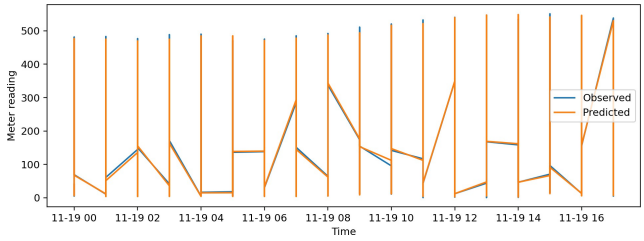


Fig. 2: Observed and predicted energy for the first 500 test observations.

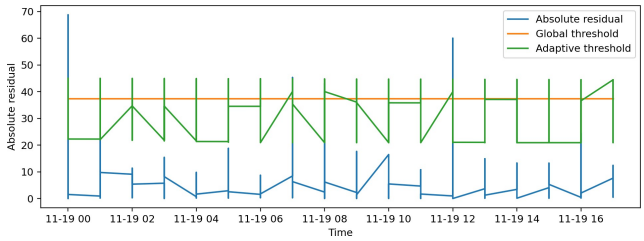


Fig. 3: Absolute prediction residual and global versus regime-aware adaptive thresholds for the first 500 test observations.

the LightGBM predictions. Figure 6 gives the overall summary, while Fig. 7 shows one observation in detail. The largest residual in the evaluated subset occurred for building 121 at 00:00 on December 4, 2016. Measured energy was 8.306, whereas the model predicted 478.183, giving an absolute residual of 469.877. This mismatch merits investigation. The SHAP plot describes the features that influenced the prediction; it cannot establish the physical cause of the mismatch.

E. Computational Analysis

Training LightGBM took approximately 1.95 seconds in the environment used for this experiment. This measurement depends on the hardware and software configuration and should not be treated as a general runtime guarantee. Inference latency and model size were not measured. Both should be evaluated on the intended building-management system before deployment.

F. Comparison of Detection Trade-offs

The methods suit different priorities. The rolling threshold performed worst on F1, while Isolation Forest achieved the highest recall and F1 but also generated the most false alarms. The proposed detector may be useful when alert precision and manageable alert volume matter most: precision rose from 0.293 for the global threshold to 0.372, and false alarms fell from 10.23 to 6.49 per 1,000 normal observations. It is not best on every metric. An application where missing anomalies is especially costly may favor a more recall-oriented method.

VIII. Limitations and Threats to Validity

The findings should be interpreted in light of several limitations. The detector flags unusual energy behavior but does not identify a physical root cause. SHAP explains how

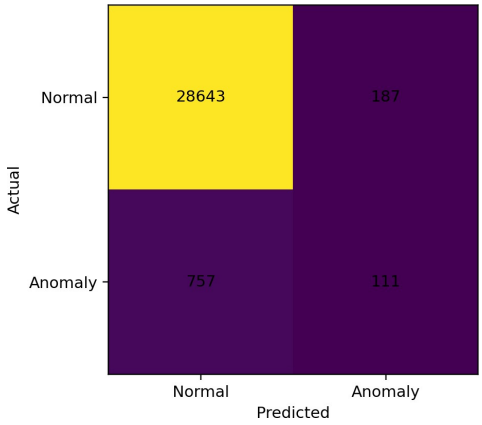


Fig. 4: Confusion matrix for the proposed regime-aware adaptive detector.

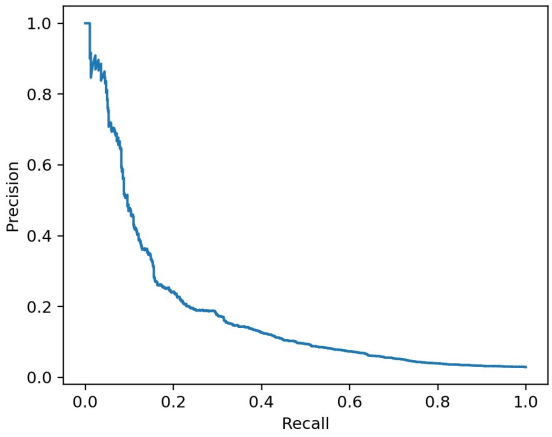


Fig. 5: Precision–recall curve for the proposed detector on the held-out test set.

input features influenced the model output; it is not a causal diagnosis. Results also depend on how representative the selected buildings and their labels are. If benchmark labels are incomplete or imperfect, measured performance may differ from what would be observed in a live facility. Regime assignment can be unreliable when the available features do not clearly distinguish operating states. Threshold settings also affect the balance between missed anomalies and false alerts, so they should be selected on validation data and reported clearly.

IX. Conclusion and Future Work

This study tested a lightweight method that combines energy forecasting with regime-aware residual thresholds. On the selected 30-building subset of LEAD1.0, LightGBM predicted energy more accurately than the persistence baseline. Compared with a global residual threshold, the proposed detector improved precision and ROC-AUC and reduced false alarms. It did not outperform every baseline: Isolation Forest achieved higher

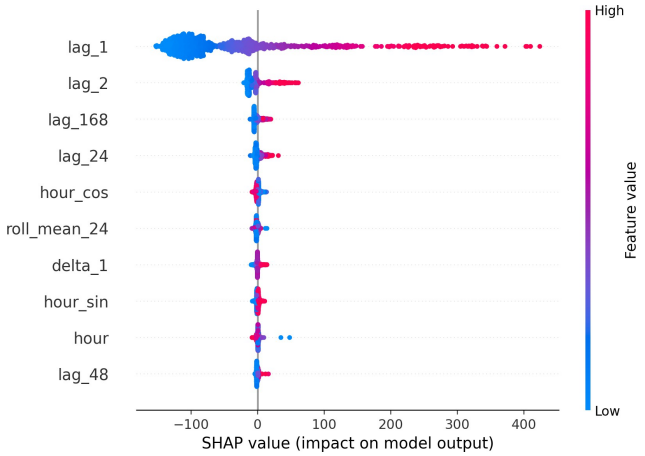


Fig. 6: SHAP summary for 1,000 sampled test observations.

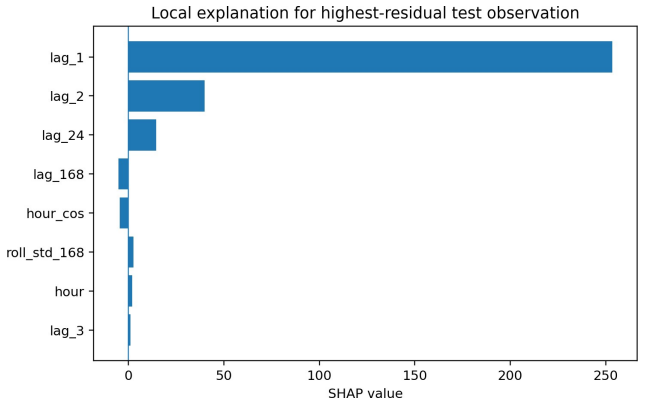


Fig. 7: Local SHAP explanation for the highest-residual test observation in the evaluated subset.

recall and F1, while the proposed method favored fewer false alerts over detecting as many labeled anomalies as possible. Whether that trade-off is acceptable depends on the priorities of the monitoring application.

The current evaluation is limited to 30 buildings and depends on the benchmark's anomaly labels; it also does not provide direct fault diagnosis. A broader evaluation should include more buildings and test whether the method transfers to buildings not seen during training. Further work could examine different seasons and building types, online adaptation to changing behavior, uncertainty estimates, and methods for identifying likely fault causes. Publishing the configuration and implementation alongside the paper would make the results easier to reproduce.

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